Final Project

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1 DATA SELECTION

1.

- Student Performance Dataset (student-por.csv, Portuguese language course)
- https://archive.ics.uci.edu/dataset/320/student+performance

2.

- Regulated Domain: Education

3.

- Observations (Rows): 649 observation

4.

- Variables (Columns): 33 total variables

5.

- G3_pass: A binary variable where 1 = Pass (G3 >= 10) and 0 = Fail (G3 < 10). This measures final course grade.
- G2_pass: A binary variable where 1 = Pass (G2 >= 10) and 0 = Fail (G2 < 10). This measures second period grade.

6.

- 2 variables: Sex, Age

7.

- Sex: Title IX of the Education Amendments of 1972
- Age: The Age Discrimination Act of 1975

2 DATA EXPLORATION

1.

- Sex: Female or Male
- Age: 15 to 22 years old

2.

Protected Class	Original Subgroup	Discrete Value
sex	F	0
sex	M	1
age	15	0
age	16	1
age	17	2
age	18	3
age	19	4
age	20	5
age	21	6
age	22	7

Table 1—Protected Class Subgroup Encoding

3.

- Chosen protected classes: Sex, Age

4.

G2_pass	Fail	Pass
sex		
F	72	311
M	73	193

Table 2—G2 Pass/Fail Status by Sex

G3_pass	Fail	Pass
sex		
F	50	333
M	50	216

Table 3—G3 Pass/Fail Status by Sex

G2_pass age	Fail	Pass
15	14	98
16	33	144
17	43	136
18	38	102
19	15	17
20	1	5
21	0	2
22	1	0

Table 4—G2 Pass/Fail Status by Age

G3_pass age	Fail	Pass
15	12	100
16	22	155
17	28	151
18	27	113
19	10	22
20	0	6
21	0	2
22	1	0

Table 5—G3 Pass/Fail Status by Age

5.

Feature Value	Count	Threshold	False Positives (%)	False Negatives (%)	Accuracy (%)	F1
African-American	5338	0.66	19.1	15.9	65.0	0.68
Caucasian	3359	0.62	8.2	26.8	65.0	0.46
Hispanic	739	0.63	6.2	26.0	67.8	0.35
Other	493	0.4	6.5	25.8	67.7	0.43
Asian	37	0.38	13.5	21.6	64.9	0.24
Native American	34	0.92	0.0	32.4	67.6	0.52

Figure 1—Sex vs G3_Pass

Feature Value	Count	Threshold	False Positives (%)	False Negatives (%)	Accuracy (%)	F1
African-American	5338	0.85	8.0	33.3	58.8	0.49
Caucasian	3359	0.62	8.2	26.8	65.0	0.46
Hispanic	739	0.63	6.2	26.0	67.8	0.35
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Figure 2—Sex vs G2_Pass

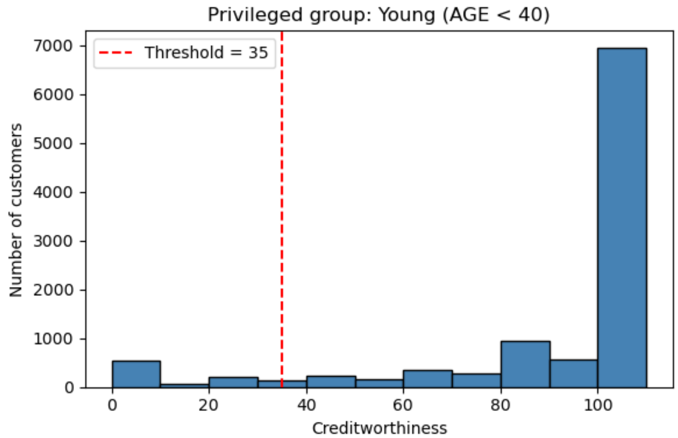

Figure 3—Age vs G3_Pass

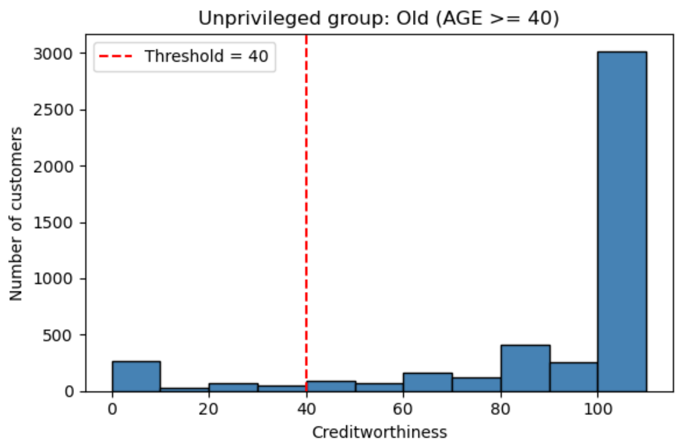

Figure 4—Age vs G2_Pass

3 FAIRNESS METRIC SELECTION AND MITIGATING BIAS

1.

- Sex privileged group: Female
- Sex unprivileged group: Male
- Age privileged group: Students of age <=17
- Age unprivileged group: Students of age >= 18

2.

Protected class	Outcome	Statistical Parity Difference	Disparate Impact
sex	G2_pass	-0.0864	0.8935
sex	G3_pass	-0.0574	0.9340
age	G2_pass	-0.1116	0.8619
age	G3_pass	-0.0775	0.9107

Table 6—Fairness Metrics Across Protected Classes and Outcomes

3.

- Selected pre-processing bias mitigation algorithm: Reweighting on sex group and G3_pass

4.

Protected class	Outcome	Statistical Parity Difference	Disparate Impact
sex	G2_pass	-0.0422	0.9467
sex	G3_pass	0.0000	1.0000
age	G2_pass	-0.1164	0.8558
age	G3_pass	-0.0820	0.9057

Table 7—Statistical Parity Difference and Disparate Impact by Demographic

4 MITIGATING BIAS

Stage	Disparate Impact	Change compared to previo
Original Dataset	0.9340	NA
After Transforming Dataset	1.0000	Positive change
After Training Classifier on Original Dataset	0.9923	Minimal negative change
After Training Classifier on Transformed Dataset	1.0008	Positive change

Table 8—Sex vs G3_pass — Disparate Impact

Stage	Statistical Parity Difference	Change compared
Original Dataset	-0.0574	NA
After Transforming Dataset	0.0000	Positive change
After Training Classifier on Original Dataset	-0.0075	Minimal negative
After Training Classifier on Transformed Dataset	0.0008	Positive change

Table 9—Sex vs G3_pass — Statistical Parity Difference

Stage	Disparate Impact	Change compared to previo
Original Dataset	0.9107	NA
After Transforming Dataset	0.9057	Negative change
After Training Classifier on Original Dataset	0.9376	Positive change
After Training Classifier on Transformed Dataset	0.9168	Negative change

Table 10—Age vs G3_pass — Disparate Impact

Stage	Statistical Parity Difference	Change compared
Original Dataset	-0.0775	NA
After Transforming Dataset	-0.0820	Negative change
After Training Classifier on Original Dataset	-0.0611	Positive change
After Training Classifier on Transformed Dataset	-0.0815	Negative change

Table 11—Age vs G3_pass — Statistical Parity Difference

5 ANALYSIS

I am a team of one

REFERENCES

[1] Blow, C. H., Qian, L., Gibson, C., Obiomon, P., & Dong, X. (2023). "Comprehensive validation on reweighting samples for bias mitigation via AIF360." *arXiv preprint arXiv:2312.12560*. URL: https://arxiv.org/abs/2312.12560.